

HAMZA SHAHZAD

AI Engineer | Python Developer | Machine Learning Engineer

Pakistan | hamzashahzad78374@gmail.com | +92 320 7074141

github.com/hamza01055 | www.linkedin.com/in/hamza-shahzad-667602355

PROFESSIONAL SUMMARY

BS Artificial Intelligence graduate (2026) with hands-on experience building AI applications using Python, FastAPI, LangChain, LangGraph, and Retrieval-Augmented Generation (RAG). Built multiple end-to-end AI projects, including multi-agent systems, RAG-based document chat platforms, computer vision applications, and a cross-platform AI productivity app. Passionate about learning, solving real-world problems, and contributing to production AI systems.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, SQL, C++

CS Fundamentals: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP)

AI & Machine Learning: Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, Prompt Engineering

LLM & Agentic AI: LangChain, LangGraph, OpenAI API, Hugging Face, Ollama, RAG, Vector Databases

Backend: FastAPI, Flask, Django

Frontend & Mobile: React, Next.js, Tailwind CSS, Flutter (Riverpod, Hive)

Databases: PostgreSQL, MongoDB, Redis, FAISS, ChromaDB

Cloud & Tools: AWS, Docker, Git, GitHub, Linux, Vercel, Render

PROFESSIONAL EXPERIENCE

Freelance AI Developer — Self-Employed

2025 – Present

- Developed AI applications using Python and FastAPI for real-world use cases.
- Built Retrieval-Augmented Generation (RAG) applications using LangChain and vector databases (FAISS, ChromaDB).
- Created multi-agent AI workflows with LangGraph for research, analysis, and report generation.
- Developed REST APIs and integrated LLM APIs (OpenAI) into client workflows.
- Used Docker and Git for development, version control, and deployment.

FEATURED PROJECTS

IntelliVault | github.com/hamza01055/IntelliVault

Flutter · FastAPI · LangChain · PostgreSQL · Redis · Docker

- AI productivity platform enabling users to chat with documents via LLM-powered RAG, capture intelligent notes, and organize tasks.
- Built a Flutter mobile app (Riverpod, Hive, clean architecture) with JWT-secured FastAPI backend.

AI Business OS | github.com/hamza01055/multi-agent-business-os

Python · FastAPI · LangGraph · Docker · PostgreSQL · React

- Built a multi-agent AI platform with document chat using RAG.
- Added OCR and Whisper speech-to-text features; developed REST APIs with FastAPI.

Multi-Agent Financial Analyst | github.com/hamza01055/multi-agent-financial-analyst

Python · LangGraph · FastAPI

- Built AI agents for financial research with automated investment report generation and RAG-based document retrieval.

Smart City Detection System | github.com/hamza01055/smartcity-ai

YOLOv8 · OpenCV · FastAPI · React

- Detected city issues from images using YOLOv8, with a FastAPI prediction backend and React dashboard.

Portfolio Website | github.com/hamza01055/Portfolio-Website

React · Framer Motion · Tailwind CSS · Three.js

- Designed and built a modern animated portfolio showcasing AI projects with 3D graphics.

EDUCATION

BS Artificial Intelligence — Islamia University of Bahawalpur

2022 – 2026

LANGUAGES

Urdu: Native **English:** Professional Working Proficiency **French:** Beginner (A2)